

# Canada-U.S. Trade Shock Observatory

## 2025 Section 232 core steel

A reproducible account of U.S. imports for consumption from Canada, built from official Census records and matched to the 2025 tariff schedule.

Ridoy Roy

| 2025 IMPORT VALUE | ANNUAL CHANGE | 2025 CALCULATED DUTY |
|-------------------|---------------|----------------------|
| \$4.51B           | -36.8%        | \$1.07B              |

The value of Canadian core-steel imports in scope was 36.8% lower in 2025 than in 2024. Declines were larger in later policy windows, although this comparison alone does not establish causality.

RELEASED SEPTEMBER 2026 / VALIDATION PASSED / 108 ARCHIVED API RESPONSES

# 1. Scope and methodology

The observatory measures monthly U.S. imports for consumption of Canadian core steel. Product scope follows the original Section 232 steel ranges and excludes later derivative articles.

| Dimension        | Initial release specification                                          |
|------------------|------------------------------------------------------------------------|
| Trade flow       | U.S. imports for consumption from Canada (Census country code 1220)    |
| Coverage         | January 2024 through December 2025                                     |
| Product universe | 746 HTS10 lines mapped to 168 HS6 codes                                |
| Outcomes         | Import value, dutiable value, calculated duty; quantity where reported |
| Data grain       | HS6-country-month for full scope; HTS10-country-month for pilot        |
| Source handling  | Exact JSON responses retained; release manifest pins SHA-256 hashes    |

## Policy timeline

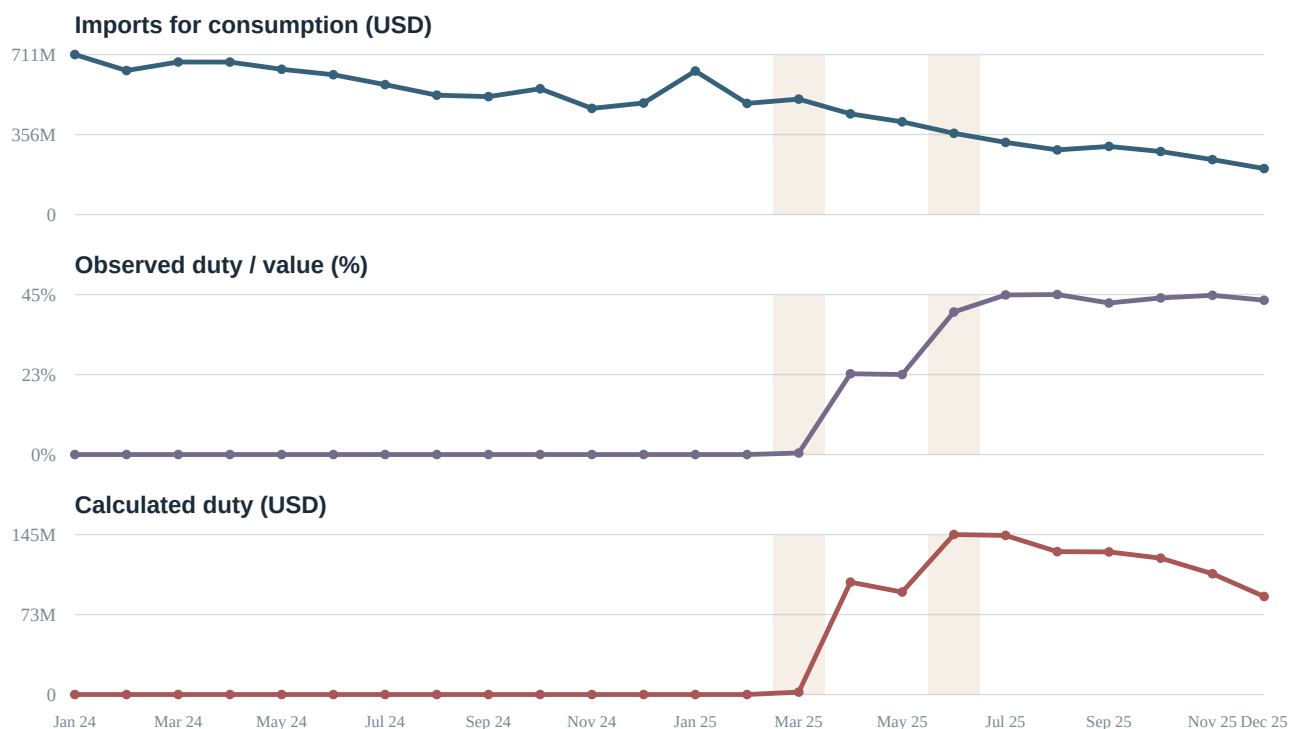
|                        |     |                              |
|------------------------|-----|------------------------------|
| Before Mar. 12, 2025   | 0%  | Canadian exemption in effect |
| Mar. 12 - Jun. 3, 2025 | 25% | Exemption terminated         |
| From Jun. 4, 2025      | 50% | Additional rate increased    |

March and June are transition months. The panel reports start-of-month, end-of-month, and calendar-day-weighted statutory rates, but deliberately leaves the clean-month treatment rate blank. Day weighting is descriptive and does not assume imports were evenly distributed across days.

Census returns both a rate-provision total row (RP='-') and its detailed breakdown. The production panel uses the total row and validates this behavior to prevent double-counting.

## 2. Full core-steel results

Import value generally declined during 2025, while reported calculated duty increased after the March and June policy changes. The duty-to-value series divides Census calculated duty by total import value.



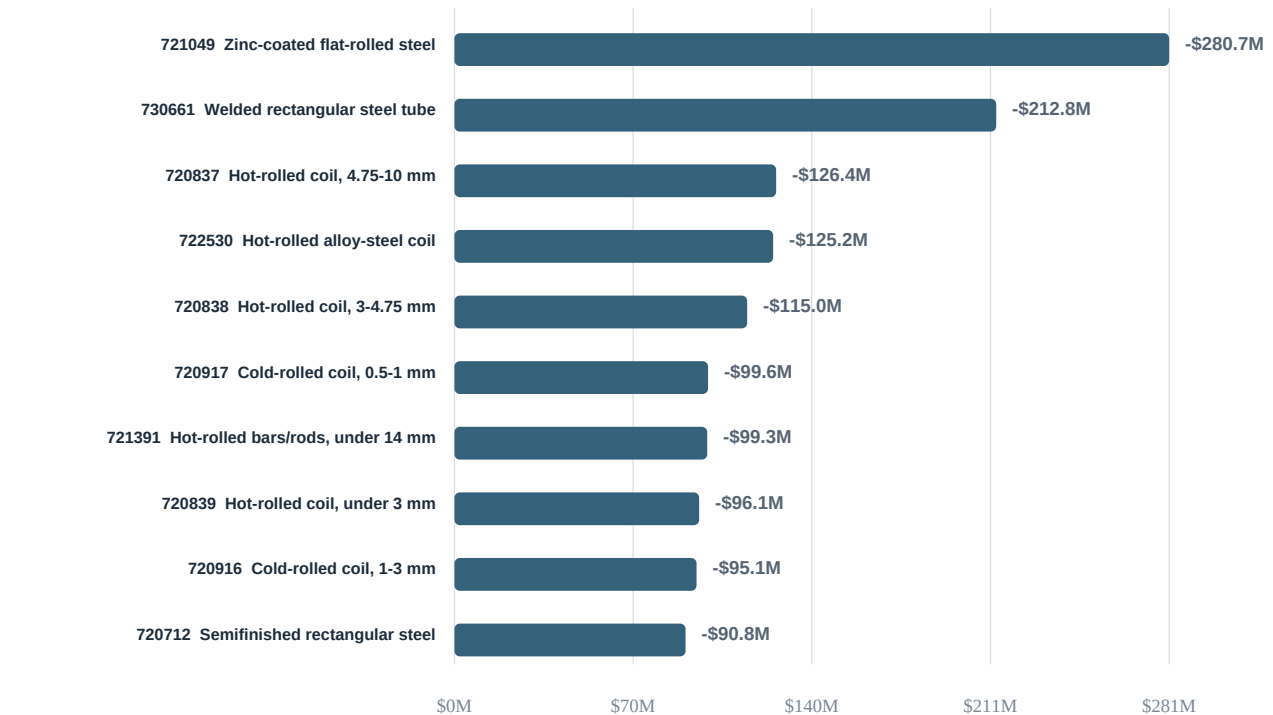
Source: U.S. Census Bureau International Trade API; author's calculations. Note: Warm shading marks March and June 2025 transition months.

| 2025 clean window     | Import value | vs. same 2024 months | Duty / value |
|-----------------------|--------------|----------------------|--------------|
| Pre-tariff: Jan.-Feb. | \$1.132B     | -16.2%               | 0.0%         |
| 25%: Apr.-May         | \$859.8M     | -35.0%               | 22.7%        |
| 50%: Jul.-Dec.        | \$1.640B     | -48.1%               | 44.3%        |

**Key interpretation:** The year-over-year decline is larger in each successive clean policy window. Prices, demand, seasonality, inventory decisions, product substitution, and advance shipments may also matter. A causal estimate would require a comparison group and a specified counterfactual.

### 3. Products contributing most to the decline

The chart ranks HS6 products by the reduction in annual import value from 2024 to 2025. Together, the top ten account for a substantial share of the aggregate decline, led by zinc-coated flat-rolled steel (HS6 721049) and welded rectangular tubing (HS6 730661).

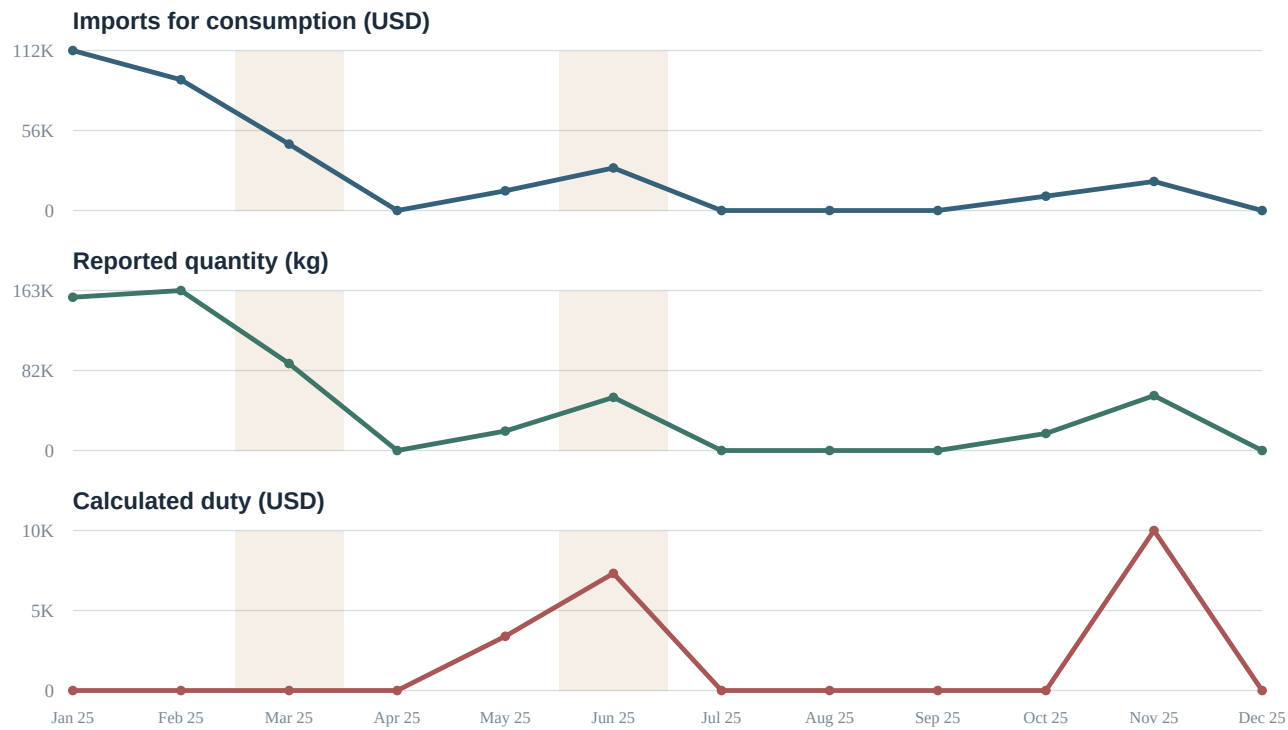


| Rank | HS6    | Product                           | 2024 value | 2025 value | Change    | Share |
|------|--------|-----------------------------------|------------|------------|-----------|-------|
| 1    | 721049 | Zinc-coated flat-rolled steel     | \$817.3M   | \$536.6M   | -\$280.7M | 10.7% |
| 2    | 730661 | Welded rectangular steel tube     | \$447.9M   | \$235.1M   | -\$212.8M | 8.1%  |
| 3    | 720837 | Hot-rolled coil, 4.75-10 mm       | \$241.7M   | \$115.3M   | -\$126.4M | 4.8%  |
| 4    | 722530 | Hot-rolled alloy-steel coil       | \$233.2M   | \$108.0M   | -\$125.2M | 4.8%  |
| 5    | 720838 | Hot-rolled coil, 3-4.75 mm        | \$222.9M   | \$107.9M   | -\$115.0M | 4.4%  |
| 6    | 720917 | Cold-rolled coil, 0.5-1 mm        | \$214.1M   | \$114.5M   | -\$99.6M  | 3.8%  |
| 7    | 721391 | Hot-rolled bars/rods, under 14 mm | \$300.5M   | \$201.2M   | -\$99.3M  | 3.8%  |
| 8    | 720839 | Hot-rolled coil, under 3 mm       | \$181.4M   | \$85.3M    | -\$96.1M  | 3.7%  |
| 9    | 720916 | Cold-rolled coil, 1-3 mm          | \$219.4M   | \$124.3M   | -\$95.1M  | 3.6%  |
| 10   | 720712 | Semifinished rectangular steel    | \$171.7M   | \$80.9M    | -\$90.8M  | 3.5%  |

**Key interpretation:** The top ten products account for approximately 51.0% of the aggregate decline. This is an accounting decomposition of the observed value change, not a causal estimate of product-level tariff effects.  
Source: U.S. Census Bureau International Trade API; author's calculations.

## 4. HTS10 pilot: 7208101500

Flat-rolled iron or nonalloy steel coils, 600 mm or more wide, hot-rolled, pickled, with patterns in relief. Unlike the HS6 aggregate release, this line reports quantity in kilograms.



| Month   | Value     | Quantity (kg) | Duty     | Policy                 |
|---------|-----------|---------------|----------|------------------------|
| 2025-01 | \$111,565 | 156,594       | \$0      | No Section 232 tariff  |
| 2025-02 | \$91,190  | 163,466       | \$0      | No Section 232 tariff  |
| 2025-03 | \$46,308  | 88,864        | \$0      | Transition: 0% to 25%  |
| 2025-04 | \$0       | 0             | \$0      | 25% tariff             |
| 2025-05 | \$13,746  | 19,876        | \$3,436  | 25% tariff             |
| 2025-06 | \$29,683  | 54,317        | \$7,421  | Transition: 25% to 50% |
| 2025-07 | \$0       | 0             | \$0      | 50% tariff             |
| 2025-08 | \$0       | 0             | \$0      | 50% tariff             |
| 2025-09 | \$0       | 0             | \$0      | 50% tariff             |
| 2025-10 | \$10,000  | 17,423        | \$0      | 50% tariff             |
| 2025-11 | \$20,267  | 56,096        | \$10,134 | 50% tariff             |
| 2025-12 | \$0       | 0             | \$0      | 50% tariff             |

**Key interpretation:** Before March 12, Canada remained exempt from the additional Section 232 steel tariff. After that date, a zero for this pilot can indicate either no reported imports in the month or a reported entry with no Census calculated duty. The chart reproduces reported Census values; it does not replace them with the headline tariff rate.

## 5. Quality, limitations, and reproducibility

### Release quality controls

| Check                   | Result                                                              |
|-------------------------|---------------------------------------------------------------------|
| Automated tests         | 23 passed, including a complete offline release build               |
| Official HS6-month rows | 3,694                                                               |
| Months                  | 24                                                                  |
| Raw Census snapshots    | 108 (96 full scope; 12 HTS10 pilot)                                 |
| Snapshot integrity      | SHA-256 verified; API key excluded from metadata                    |
| Release provenance      | Manifest pins 108 Census responses, configuration, and core outputs |
| Transition handling     | March and June 2025 flagged; excluded from clean comparisons        |
| HS6 quantity            | Unavailable in aggregate API rows; stored as null, not zero         |

| WHAT THE EVIDENCE SHOWS                                                                                           | WHAT IT DOES NOT CLAIM                                                                                                |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Observed import values, calculated duty, policy timing, and the products contributing most to the annual decline. | A causal tariff effect, welfare impact, tariff pass-through, domestic output, employment effects, or trade diversion. |

### Reproduction

A fresh clone can run `python scripts/build_release.py` to download the official concordance and current Census data. Run `python scripts/verify_release.py` to verify the committed release against its manifest. The exact source-level rebuild uses `--reuse-snapshots` and requires the author's private raw archive; credentials and raw responses are not distributed through Git.

### Official sources

- U.S. Census International Trade API
- Census API variable definitions
- January 2025 Census commodity concordance
- February 2025 steel proclamation
- June 2025 steel and aluminum proclamation

### Publication note

Recommended citation: Roy, Ridoy. 2026. *Canada-U.S. Trade Shock Observatory: 2025 Section 232 Core Steel, Initial Analytical Release*. Data derived from the U.S. Census Bureau and official presidential proclamations.